

Introduction to Macroeconomics (S102016CR)

Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

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Second Part

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graph TD; A[Second Part] --> B[Financial Economics]; A --> C[Monetary Economics]; B --> D["The Financial System and closed economy equilibrium (chapter 4)"]; C --> E["The Monetary System (Chapter 5)"]; C --> F["Money growth and Inflation (Chapter 6)"];
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Financial Economics

The Financial System
and closed economy
equilibrium
(chapter 4)

Monetary Economics

The Monetary
System
(Chapter 5)

**Money growth
and Inflation
(Chapter 6)**

Week 6:

Money growth and Inflation

- This is the second chapter in a two-chapter sequence dealing with monetary markets. In week 5 we explained what money is and how central banks (try to) control the quantity of money in their economies.
- This week, we establish the relationship between the growth rate of money and the inflation rate in the long run.
- We will see numerous economic costs from high inflation but no consensus on the importance of these costs when inflation is moderate.
- This chapter closes the second part of the course on the monetary and financial systems
- In the fourth part of the course, we will see the relationship between money supply and other macroeconomic variables while keeping prices constant in the short run....

Chapter Contents

- a) Quantitative Theory of Money and Inflation
- b) The inflationist tax
- c) Economic inefficiencies associated with inflation
- d) Summary

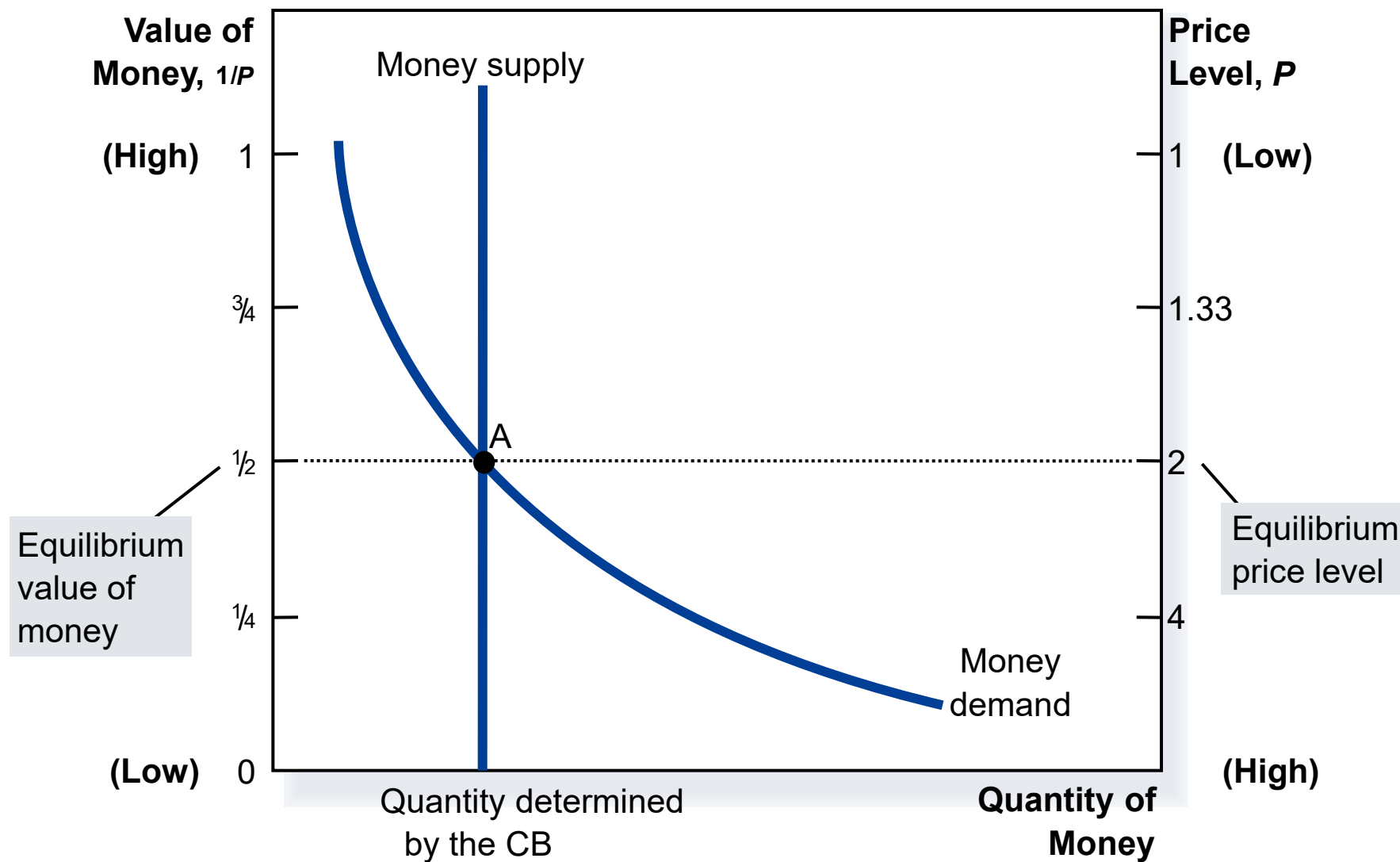
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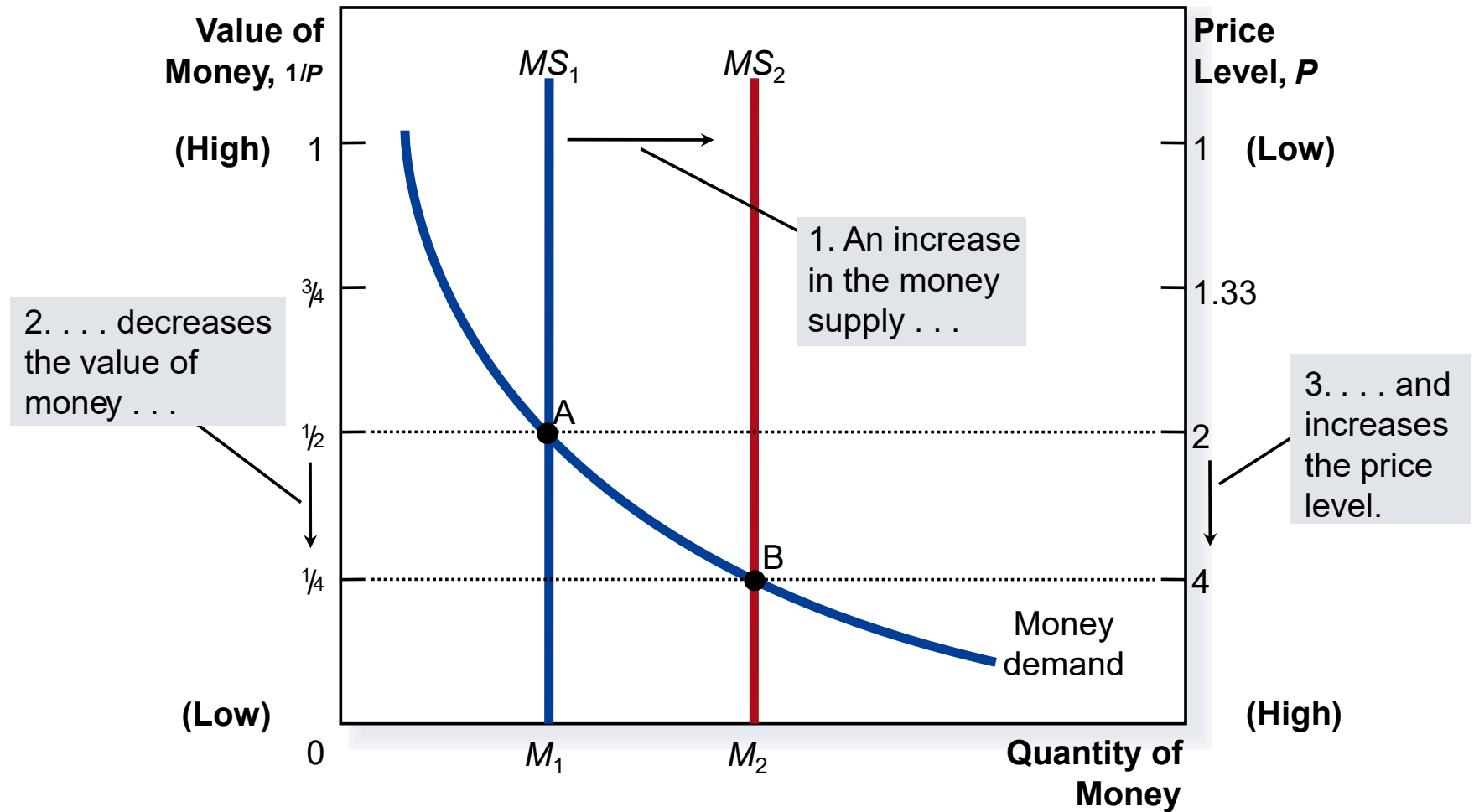
a. Quantitative Theory of Money and Inflation

- The *quantity theory of money* explains the long-run determinants of the price level and the inflation rate.
- The *money supply* is a policy variable that is (imperfectly) controlled by the central bank.
 - Through instruments such as open-market operations and refinancing interest rates, the central bank directly controls the quantity of money supplied
- *Money demand* has several determinants, including interest rates, income, and the average level of prices in the economy.
- People hold money because it is the medium of exchange.
 - The amount of money people choose to hold depends on the prices of goods and services.
- When the overall price level rises, the value of money falls. Indeed, 1CHF will buy fewer goods after prices have increased. Thus, money demand increases when prices increase (the value of money falls)

- In the long run, prices adjust to the level at which the demand for money equals the supply.



Impact of an increase in money supply on prices



- Thus, a higher money supply leads to higher prices.
- Note that this is for a given money demand function, i.e., keeping real interest rates and real income fixed.
- If real income were to change with money supply, we may not observe this impact as money demand will shift outwards, which may reverse the price increase
- But with fixed real output, when the money supply increases, there is more money chasing the same number of goods, and therefore prices will increase to clear markets.....
- In the long run, real output is fixed as the determinants of long-run real output and growth do not depend on money supply but on capital accumulation (physical or human), technology, and natural resources.
- We then say there is “money neutrality” on real variables in the long run.
- Only nominal variables (prices) will be affected in the long run
- This is the classic dichotomy (e.g., Hume) regarding the impact of money supply on nominal and real variables. Only the former will be affected.

The Quantitative Theory of Money

- shows the long-run impact of changes in the money supply on prices
- In the long run, the primary cause of inflation is the growth in the quantity of money.

The *velocity of money* refers to the speed at which the money changes hands, traveling around the economy from wallet to wallet.

$$V = (P \times Y)/M$$

where: V is velocity, P is the price level, Y is the quantity of output, and M is the quantity of money

- Rewriting the equation gives the *quantitative equation*:

$$M \times V = P \times Y$$

- The *quantitative equation* relates the quantity of money (M) to the nominal value of output ($P \times Y$).

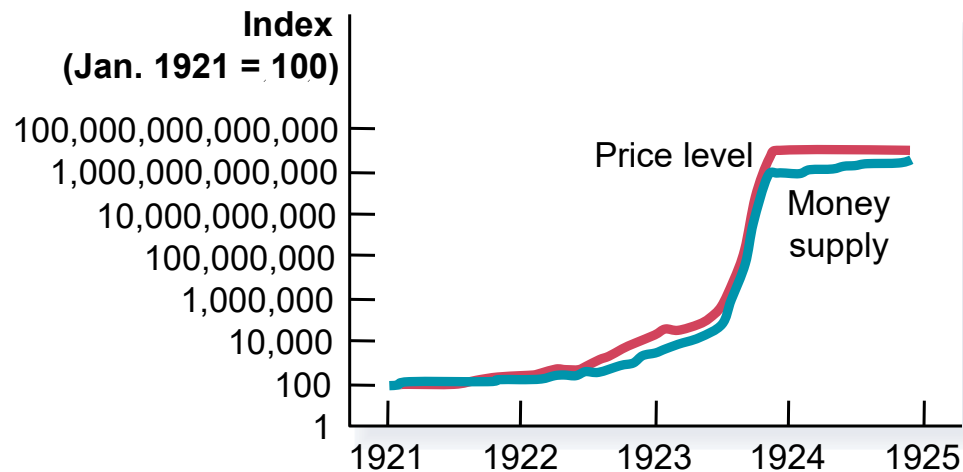
- The quantity equation shows that an increase in the quantity of money in an economy must be reflected in one of three other variables:
 - the price level must rise,
 - the quantity of output must rise, or
 - the velocity of money must fall.
- The Quantity Theory of Money assumes
 - The velocity of money is relatively stable over time.
 - In the long run, money is neutral. Money does not affect output.

• Thus

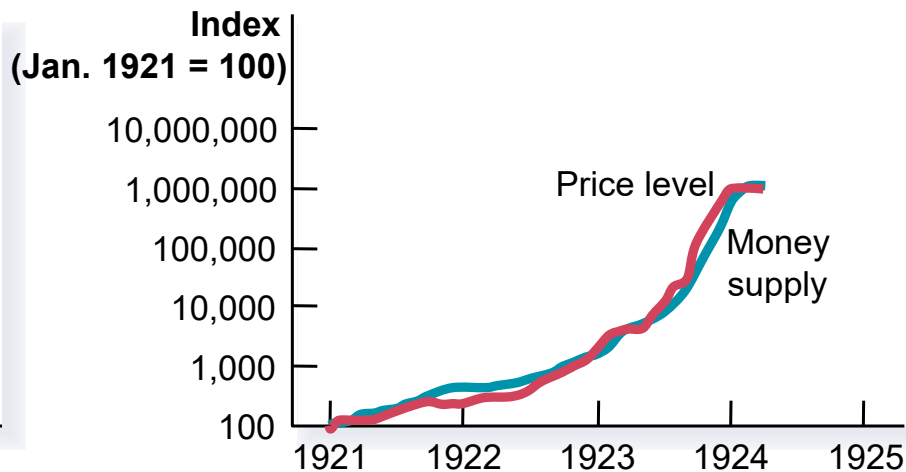
$$\Delta M \times V = \Delta P \times Y \Rightarrow \Delta M = \Delta P \times Y/V \Rightarrow \Delta M/M = \Delta P/P$$

- This prediction fits the data well during periods of hyperinflation (inflation rates above 50% per month or 13000% annually)

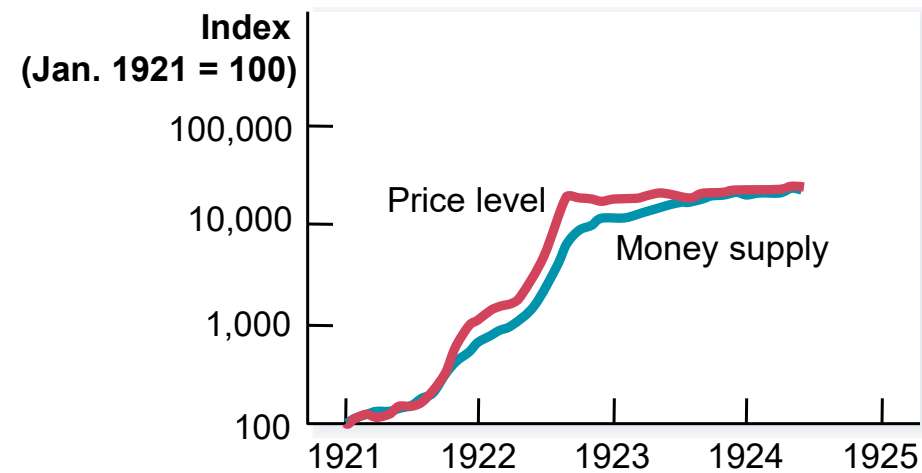
Germany



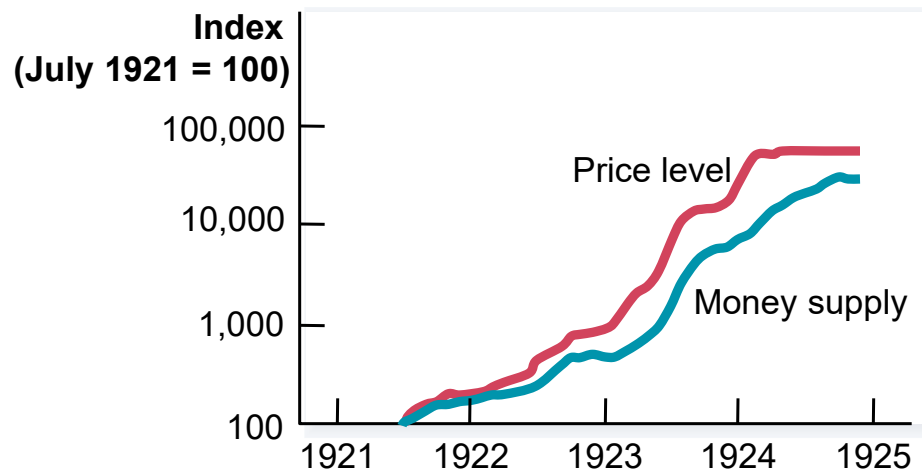
Poland



Austria

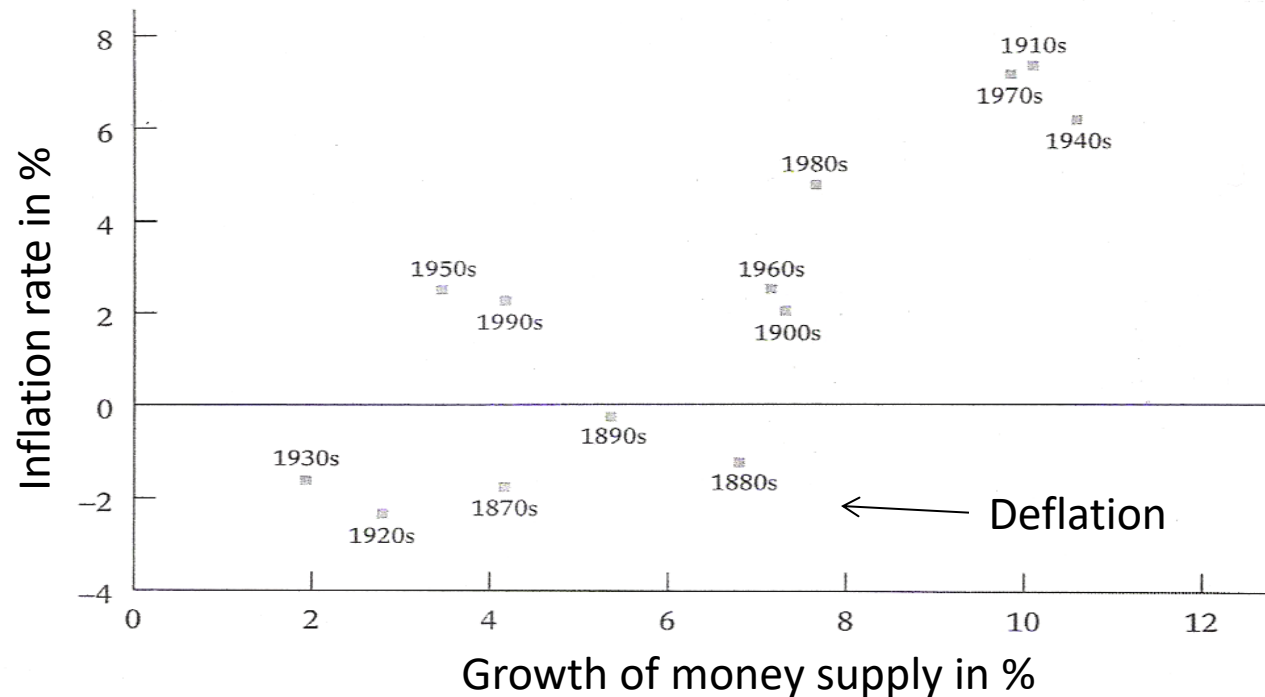


Hungary



With more moderate levels of inflation or deflation, there is also a positive correlation between inflation and growth in the money supply (in the long run)

Historical data on inflation and the growth of money supply in the United States



Source: Milton Friedman and Anna Schwartz.

Note: money refers to M2

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b. The inflationist tax

- But why do Central Banks print money if this eventually creates inflation and erodes the value of money?
- To pay for government expenditure, especially when alternatives are not available (higher taxes or issuing bonds)
- All hyperinflation episodes have a similar starting point: governments have a growing budget deficit, increasing interest rate spreads on their bonds, and the capacity to tax is low (because the economy is in recession already).
- The government then asks the CB to print money to finance its spending needs.
- This reduces the value of money as prices increase, and therefore it is a tax on everyone else who holds cash. We call this an inflationist tax.
- When the government raises revenue by printing money, it is said to levy an *inflation tax*.
- An inflation tax is like a tax on everyone who holds money.
- The inflation ends when the government institutes fiscal reforms such as cuts in government spending.

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c. Economic inefficiencies associated with inflation

- Is inflation per se a fall in Purchasing Power?
 - No, inflation *does not* reduce people's purchasing power.
 - Inflation implies that all prices are, on average, increasing, including the price of labor and capital. Therefore, wages and interest on capital are also increasing
- But purchasing power will fall because of economic inefficiencies associated with inflation

Shoe leather costs

- are the resources wasted when inflation encourages people to reduce their money holdings. Inflation reduces the value of money, so people have the incentive to minimize their cash holdings.
- Less cash requires frequent bank trips to withdraw money from interest-bearing accounts. Today you could do this online, so no big deal?
- Well, the actual cost of reducing your money holdings is the time and convenience you must sacrifice to keep less money on your sight deposits and more on your interest-bearing accounts.

Menu costs

- are the costs of adjusting prices.
- During inflationary times, updating price lists and other posted prices is necessary. This is a resource-consuming process that takes away from other productive activities.
- Online menus have made this easier, but still costly
- Consumers may not appreciate the rapid changes in prices

Misallocation of resources

- Inflation distorts relative prices.
- Consumer decisions are distorted, and markets are less able to allocate resources to their best use.

The redistributive impact hurts savings and, therefore, long-run growth

- Unexpected inflation redistributes wealth away from savings
- Lower incentives to save reduces loanable funds available and therefore investment

Confusion and inconvenience

- Inflation causes money at different times to have different real values.
- Therefore, with rising prices, it is more difficult to compare real revenues, costs, and profits over time, raising uncertainty and making investment more difficult.
- Less investment reduces growth

Inflation-induced tax distortions

- Inflation exaggerates the size of capital gains and increases the tax burden on this type of income.
- The nominal interest earned on savings is treated as income for income tax purposes, even though part of the nominal interest rate merely compensates for inflation.
- Consequently, the after-tax real interest rate falls when inflation rises, making saving less attractive.
- With progressive taxation, capital gains are taxed even more heavily, making this problem even worse.

Example of how inflation increases the burden of taxation, and will reduce savings hurting long-run growth

	Economy A (price stability)	Economy B (inflation)
Real interest rate	4%	4%
Inflation rate	0	8
Nominal interest rate (real interest rate + inflation rate)	4	12
Reduced interest due to 25 percent tax (.25 × nominal interest rate)	1	3
After-tax nominal interest rate (.75 × nominal interest rate)	3	9
After-tax real interest rate (after-tax nominal interest rate – inflation rate)	3	1

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d. Summary

- The overall level of prices in an economy adjusts to bring money supply and demand into balance.
- When the central bank increases the money supply, it causes the price level to rise.
- Persistent growth in the quantity of money supplied leads to continuing inflation.
- The principle of money neutrality asserts that changes in the quantity of money influence nominal variables but not real variables.
- A government can pay for its spending simply by printing more money.
- This can result in an “inflation tax” and hyperinflation.
- According to the Fisher effect, when the inflation rate rises, the nominal interest rate rises by the same amount, and the real interest rate stays the same (money neutrality implies that monetary phenomena do not affect real variables in the long run).
- Many people think that inflation makes them poorer because it raises the cost of what they buy. This view is a fallacy because inflation also raises nominal incomes.

- Economists have identified six costs of inflation:
 - Shoe leather costs
 - Menu costs
 - Increased variability of relative prices
 - Unintended tax liability changes
 - Confusion and inconvenience
 - Arbitrary redistribution of wealth